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Weakly Supervised RGBT salient object detection via SAM-Guided Label Optimization and Progressive Cross-modal Cross-scale Fusion

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ABSTRACT

Current fully supervised RGB-Thermal salient object detection (RGBT SOD) methods rely on labor-intensive pixel-wise annotations. This work explores weakly supervised RGBT SOD using scribble annotations to reduce the annotation cost. Existing scribble-supervised methods mainly rely on pseudo-labels, which often suffer from issues such as information redundancy, incomplete objects, or inaccurate boundaries. Inspired by the advanced Segment Anything Model (SAM), we propose a two-stage SAM-Guided Label Optimization method to obtain accurate pseudo-labels. The first stage employs the prompt-based segmentation ability of SAM to generate initial masks from RGB and thermal images, while the second stage refines these initial masks using the zero-shot segmentation capability of SAM and complementary information from RGB and thermal modalities. Moreover, existing multi-modal fusion methods might not fully synergize the interactions between channel and spatial dimensions, and they often neglect effective cross-scale feature collaboration during the multi-modal fusion. To address this, we propose the Progressive Cross-modal Cross-scale Fusion Unit (PCCFU), which fuses same-level multi-modal features while progressively integrating higher-level features. PCCFU consists of the Dual Cross-attention Fusion Module and the Cross-scale Aggregation Module. The former facilitates synergistic interactions across both the channel and spatial dimensions, while the latter interacts individually with each higher-level feature during multi-modal feature fusion. Extensive experiments indicate that our method outperforms most fully supervised RGBT SOD approaches and surpasses the previous state-of-the-art weakly supervised method, achieving average improvements of 3.4% in mean F-measure (F_{β}^{avg}) and 21.5% in mean absolute error (MAE) across three benchmark datasets. The code is available at: https://github.com/tzz-ahu.

1. Introduction

Salient Object Detection (SOD) aims to automatically identify the most prominent regions or objects in images and has a wide range of applications, including image segmentation [1,2], visual recognition [3,4] and autonomous driving [5,6]. Traditional SOD approaches primarily operate on RGB images, which often face significant challenges under adverse conditions such as bad weather, poor lighting, or complex backgrounds. In contrast, thermal infrared images capture the thermal radiation and temperature distribution of objects, offering a complementary perspective in complex scenarios where RGB images are less effective. Recently, combining RGB and thermal images [7–11] for detection has significantly improved the accuracy of SOD.

Although fully supervised RGBT SOD methods [9,10,12–15] have made significant advancements, they rely on pixel-wise annotations, which is a major limitation. Creating such annotations requires labeling each pixel, which is time-consuming and labor-intensive. In contrast, scribble annotations, which only need to outline contours of objects, can alleviate this burden substantially. As shown in Fig. 1(*column d*), green lines represent the foreground, while blue lines indicate the background. Given the flexibility and convenience of scribble

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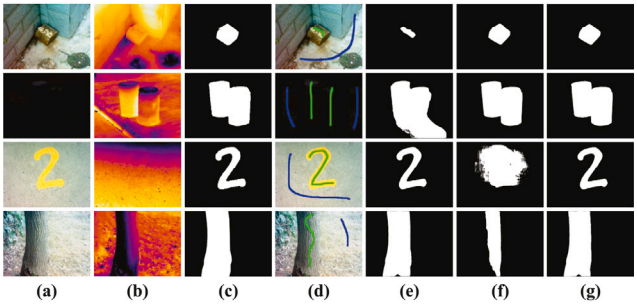

Fig. 1. Visualization of training examples and various types of labels. (a) RGB images. (b) Thermal images. (c) Ground truth. (d) Scribble annotations. (e) Initial RGB pseudo-labels. (f) Initial Thermal pseudo-labels. (g) Final pseudo-labels.

annotations, we employ them as supervisory information. However, scribble-supervised RGBT SOD methods still face several challenges.

First, scribble annotations lack detailed object structures and boundary information. Most existing scribble-supervised methods typically adopt a two-stage training approach. First, pseudo-labels are generated from scribbles, which are subsequently utilized for model training. To generate pseudo-labels, some methods [16,17] incorporate edge features for more accurate detection of object structures. Some methods [11,18,19] adopt the superpixels to expand the foreground in scribble annotations. However, the pseudo-labels generated by these methods may suffer from issues such as information redundancy, incomplete targets, or inaccurate boundaries, leading to a significant performance gap compared to fully supervised methods. Recently, the Segment Anything Model (SAM) [20] has demonstrated powerful zero-shot segmentation and prompt-based segmentation capabilities. Hence, some weakly supervised methods utilize SAM for pseudo-label generation. For example, Wang et al. [21] train an image-level classifier to generate class activation maps [22] based on image-level labels, which are subsequently used as prompts for SAM to produce pseudo-labels. Wang et al. [23] first fuse features from multiple self-supervised models [24] to generate the first type of pseudo-label. Then, they use the SAM model to generate the second and the third types of pseudo-labels based on point prompts and box prompts, respectively. Finally, the three types of pseudo-labels are fused to produce the final result. Though existing SAM-based methods have achieved performance on par with fully supervised ones, they have not been explored in the field of multi-modal SOD and their label optimization strategies are not suitable for the fusion of multi-modal pseudo-labels.

Second, leveraging complementary information is critical for multi-modal SOD tasks. Some methods [9,10,12] use simple fusion techniques like concatenation, addition, or multiplication, leading to the fusion process being dominated by one modality, as well as suppressing valid information from the other modality. Some methods [13,14] employ a cross-attention mechanism [25] in the spatial dimension, followed by the Squeeze-and-Excitation (SE) [26] mechanism to select relevant channel information. However, the SE mechanism may overly suppress other potentially useful information. Pang et al. [15] use cross-attention mechanisms in both spatial and channel dimensions for parallel feature fusion. However, this may fail to capture deeper correlations between the two dimensions. Moreover, existing methods typically focus on multi-modal feature fusion at the same level, overlooking the importance of cross-level interactions.

Unlike previous SAM-based weakly supervised methods for single-modality [21,23,27], which may lack effective optimization strategies for pseudo-labels, we propose a two-stage SAM-Guided Label Optimization (SGLO) method to generate high-quality pseudo-labels for RGBT SOD. Our approach consists of the initial pseudo-label generation and the pseudo-label optimization. Initially, we sample points from both foreground and background scribbles, and then provide these prompts and original images to SAM to generate initial pseudo-labels for each modality. However, these labels may be inaccurate due to image defects and the limitations of SAM, as illustrated in Fig. 1(columns e, f). Therefore, we propose a SAM-based pseudo-label optimization method in the second stage. The original images are input into SAM to generate a series of segmentation masks, and the most relevant masks are selected to enhance the initial pseudo-labels. Finally, we combine the complementary advantages of the two modalities and merge the optimized and the initial pseudo-labels from both modalities to acquire the final pseudo-labels. As shown in Fig. 1(column g), the pseudo-labels produced by our SGLO method closely resemble manually annotated ground truth.

Unlike previous multi-modal fusion methods [9,10,12–15] that insufficiently synergize the interactions of multi-modal features across different dimensions and treat multi-modal and cross-scale fusion separately, we propose the Progressive Cross-modal Cross-scale Fusion Unit (PCCFU), which fuses multi-modal features at the same level while progressively integrating higher-level features. The PCCFU consists of two key components: the Dual Cross-attention Fusion Module (DCFM) and the Cross-scale Aggregation Module (CSAM). The DCFM initially explores correlations between multi-modal channel information, providing important guiding information for subsequent interactions in the spatial dimension. This effectively coordinates the interactions between channel and spatial dimensions. Meanwhile, the CSAM progressively facilitates cross-scale interactions between the fused features from the DCFM of the same PCCFU and those from higher-level DCFMs, enhancing information propagation and refinement.

Here we summarize the major contributions of our work to the scribble-supervised RGBT SOD task.

- We propose a two-stage SAM-Guided Label Optimization (SGLO) strategy to obtain high-quality pseudo-labels from scribble annotations, which are close to actual ground truth. SGLO utilizes SAM's prompt-based segmentation ability to generate initial masks, then uses SAM's zero-shot segmentation ability and the complementary information of two modalities to refine the initial masks.
- We propose the Progressive Cross-modal Cross-scale Fusion Unit (PCCFU), which comprises a Dual Cross-attention Fusion Module and a Cross-scale Aggregation Module. PCCFU fuses multi-modal features at the same level through collaborative interactions across different dimensions and progressively integrates all higher-level features, simultaneously leveraging both multi-modal and multi-scale information.
- Comprehensive experiments reveal that our proposed method not only achieves state-of-the-art performance among weakly supervised methods, but also is comparable to top-performing fully supervised methods.

2. Related work

2.1. Segment anything model

The recently proposed Segment Anything Model (SAM) [20] is a foundational visual model known for its advanced segmentation capabilities. SAM can automatically segment all objects in an image, and can segment specific objects based on user prompts. Since being released, SAM has attracted significant attention and generated impressive results in various weakly supervised tasks. For instance, Wang et al. [23] are the first to leverage SAM to generate pseudo-labels for weakly supervised SOD. They propose two prompt generation methods, that are, one based on foreground scribble points and the other based on bounding boxes from self-learning pseudo-labels. Liu et al. [27] propose a SAM-based method for weakly supervised concealed object segmentation, which generates pseudo-labels from sparse scribbles and enhances the quality through multi-view augmentation and entropy weighting.

For weakly supervised semantic segmentation task, Kweon et al. [28] use the region prototype contrast method [29] to compare the automatic segmentation results from SAM with the classifier features, thereby transferring SAM's segmentation knowledge to the classifier at the feature level and improving the quality of CAMs. The CAMs are then converted into point prompts for SAM to generate category-specific masks. Zhang et al. [30] address SAM's generalization issues in downstream segmentation tasks by proposing a source-free self-training architecture, using three types of weak supervision, including bounding boxes, points, and coarse segmentation masks. This architecture combines the anchor regularization and the low-rank fine-tuning to enhance the generalization capability of SAM.

Motivated by the capabilities of the prompt-based segmentation and zero-shot generalization of SAM, we aim to apply SAM to the scribble-supervised RGBT SOD methods. According to our investigation, this research area is relatively unexplored. We expect to utilize the advantages of SAM to enhance the performance of weakly supervised RGBT SOD, narrowing the gap between them and fully supervised methods.

2.2. Scribble-supervised salient object detection

Utilizing weak annotations significantly reduces the annotation burden compared to pixel-level annotations. Common weak annotations include bounding boxes [31], image captions [32], points [33], and scribbles [11,16,17,34,35]. Among these, scribble annotations strike a good balance between annotation speed and performance. Liu et al. [11] expand sparse scribble annotations through superpixel segmentation on the images of two modalities, and design a prediction module to generate pseudo-labels. Zhang et al. [16] introduce a scribble-supervised SOD method for the first time. They leverage boundary information and propose an enhancement strategy that employs DenseCRF [36] to refine the initial predictions. Xu et al. [17] present a training method termed Predictive Consistency, involving the collaborative training on both a teacher model and a student model. They also introduce a manually assisted scribble enhancement strategy. Yu et al. [34] propose a one-round end-to-end training approach, and develop a local coherence loss and a saliency structure consistency loss. Liet al. [35] integrate a tree-energy loss based on the minimum spanning tree into their network, enabling predictions during training. Additionally, they introduce a conditional multi-modal variational autoencoder to further enhance their predictions.

The majority of these methods utilize a two-stage training strategy. However, compared to fully supervised methods, they exhibit a considerable performance gap, indicating the discrepancies between generated pseudo-labels and actual ground truth. Different from previous work, we develop a two-stage SAM-Guided Label Optimization method (SGLO). SGLO effectively employs SAM, scribble annotations, and multi-modal information to generate pseudo-labels that are close to pixel-wise ground truth.

2.3. Multi-model feature fusion methods

In multi-modal SOD tasks, a key challenge is to design an effective feature fusion strategy. Feature fusion strategies can be categorized into three primary types: early fusion [7,8], late fusion [37,38] and middle fusion [9,10,13,15,39–44]. Early fusion merges multi-modal data at the input layer, without considering the importance of the individual features. Late fusion independently generates saliency maps for each modality before merging them, preserving unique features but potentially lose intrinsic relationships between modalities. Middle fusion is the predominant fusion method, integrating modalities during feature extraction.

For example, Liu et al. [9] compute a common spatial attention map for two modalities using element-wise multiplication, then adjust and enhance features through channel attention. Tang et al. [10] combine the features of both modalities through concatenation, resulting in the formation of modality-specific weights and achieving the effective fusion. Liao et al. [13] use spatial-wise cross-attention for feature interaction between modalities, followed by the SE mechanism to select crucial channel information. Pang et al. [15] utilize the parallel cross-attention in spatial and channel dimensions to aggregate multi-modal features. Chen et al. [39] employ a cross-guided strategy, where one modality guides the extraction of features from another modality, reducing modal differences between RGB and thermal features. Zhang et al. [40] use unsupervised clustering to extract saliency prototypes, enhancing the auxiliary modality and guiding cross-modal fusion. Jiang et al. [41] propose a mirror complementary strategy with grayscale labels that direct the RGB stream to focus on skeletal regions and the thermal stream on object contours. Wang et al. [42] generate exclusive contribution weights for modalities to guide feature processing, selecting a fusion mechanism based on these weights for complementary fusion and multi-scale integration. Wang et al. [44] utilize top-level semantic information and attention mechanisms to model the modality correlations within asymmetric windows, guiding the model to focus on salient areas and reduce the noise.

Although existing fusion methods have yielded promising results, they fail to sufficiently synergize the interactions between channel and spatial dimensions, and often overlook to explore cross-level synergistic interactions. Instead, our proposed Progressive Cross-modality Cross-scale Fusion Unit not only efficiently integrates the modality features at the same level, but also explores the correlations among features across diverse hierarchical levels.

3. The proposed approach

3.1. Overview

Together with the proposed two-stage SAM-Guided Label Optimization (SGLO) strategy, we propose a novel scribble-supervised RGBT SOD learning framework. As depicted in Fig. 2, we choose the Shunted Transformer [45] as backbone to extract features. First, for an input RGBT image pair X_i and T_i, the features derived from the two modalities across four levels are denoted as f_i^r and f_i^t ($i \in \{1,2,3,4\}$), respectively. Then, the proposed Progressive Cross-model Cross-scale Fusion Unit (PCCFU) fuses the same-level multi-modal features and the high-level multi-scale features. Within the PCCFU, the Dual Cross-attention Fusion Module (DCFM) explores inter-modal relationships through channel-wise interactions, providing important guidance for subsequent spatial-wise interactions. The Cross-Scale Aggregation Module (CSAM) performs cross-scale interactions between the features from the DCFM and those from the higher-level PCCFUs, facilitating the flow of information across different scales. Finally, the fused features from the PCCFUs are fed into the decoder to produce the saliency map. During training, the saliency map is supervised jointly by the scribble annotation and the pseudo-label generated by the SGLO strategy.

In the following sections, we will introduce the SGLO strategy in Section 3.2, the PCCFU in Section 3.3, and the decoder along with the loss function in Section 3.4.

3.2. Two-Stage SAM-Guided Label Optimization (SGLO) Strategy

Before introducing the SGLO strategy, we first define the scribble-supervised training dataset as $D = \{X_i, T_i, l_i\}_{i=1}^{N}$, where N represents the number of training samples, X_i and T_i denote the RGB and the thermal images, and l_i is the corresponding scribble annotation. In l_i, the pixel values of 1, 2, and 0 represent the foreground, background, and unknown pixels respectively. To generate high-quality pseudo-labels from sparse scribbles, our proposed SGLO strategy consists of two stages, as shown in Fig. 3. Next, we will introduce each stage in detail.

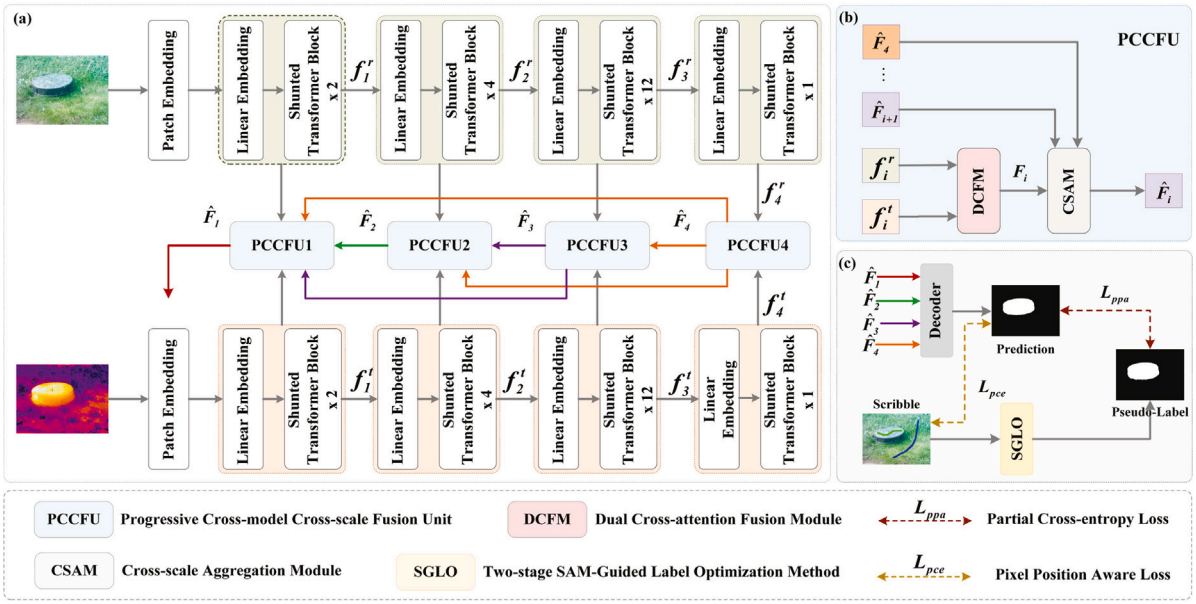

Fig. 2. Overview of our proposed network: (a) We utilize the two-stream Shunted Transformer [45] encoder to extract features. Then, the same-level features f_i^r and f_i^t are fed into the PCCFU. (b) In the PCCFU, f_i^r and f_i^t are fused through DCFM for multi-modal integration and combined with the outputs from higher-level PCCFUs using CSAM for cross-scale fusion. (c) Finally, all fused features from the PCCFU are fed into the decoder to generate the saliency map, and we use both the scribble and the pseudo-label generated from SGLO as supervisory information.

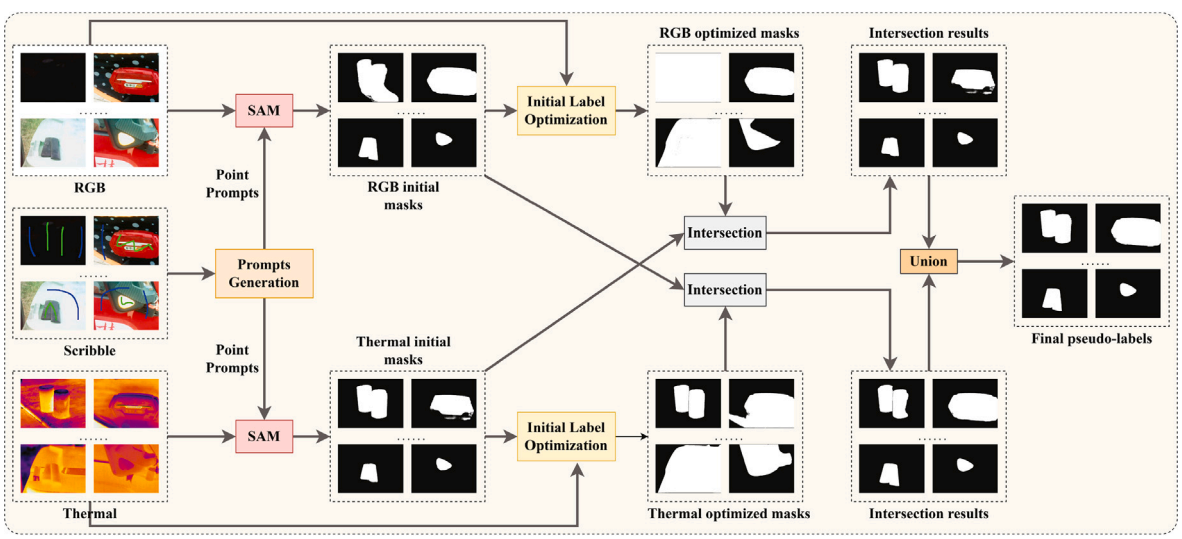

Fig. 3. Two-stage SAM-Guided Label Optimization Strategy (SGLO).

3.2.1. Initial pseudo-label generation

In the initial stage, we sample points from the scribbles as prompts for SAM to accurately segment objects in images, thereby generating initial pseudo-labels. Specifically, for a scribble annotation l_i with pixel values of 0, 1, or 2, it may contain multiple foreground and background scribble regions, as shown in Fig. 1(*column d*). To ensure uniform point sampling within each region, we apply the depth-first search (DFS) algorithm [46] to all pixels of l_i. DFS is a method for traversing a graph. It starts from a starting node and visits all directly or indirectly adjacent nodes until no more adjacent nodes are remained. In the context of an image, for each pixel of l_i, DFS visits all directly or indirectly adjacent pixels with the same value. After traversing each pixel, we can obtain multiple connected pixel regions. Among them, the regions with a pixel value of 1 represent foreground regions, and those with a pixel value of 0 represent background regions. Next, we map the points sampled from the foreground and the background scribble regions to the RGB image X_i and thermal image T_i. Using these point prompts, we obtain initial pseudo-labels P_r^{init} and P_t^{init} for the two modalities by SAM.

However, as shown in Figs. 1 and 3, the initial pseudo-labels produced by SAM might not always be reliable, particularly in challenging scenarios. Consequently, directly using them is suboptimal. Considering that SAM's zero-shot segmentation capability can automatically segment all objects in an image and generate segmentation masks, we propose a method in the second stage to optimize the initial pseudo-labels using these generated segmentation masks. In the following, we will detail this initial pseudo-label optimization approach.

3.2.2. Initial pseudo-label optimization

For a brief description, we take the optimization of the initial RGB pseudo-label P_r^{init} as an example, as illustrated in Algorithm 1. Given an input RGB image X_i, SAM can automatically segment all objects in X_i without prompts and generate a set of binary masks $\{m_i\}_{i=1}^{k}$,

Algorithm 1 Initial RGB Pseudo-label Optimization

Input: RGB image X_i, Initial RGB Pseudo-label P_r^{init}, Threshold τ
Output: Optimized RGB Pseudo-label P_r^{ref}

```
{m_i}_{i=1}^k ← SAM(X_i)                    ▷Generate a mask set
F ← {}                                      ▷Initialize a foreground mask set
for each mask m_i ∈ {m_i}_{i=1}^k do
    I ← m_i ∩ P_r^init                      ▷Intersection
    S ← min(m_i, P_r^init)                  ▷Minimum area
    r ← I/S                                 ▷Calculate ratio
    if r > τ then
        F ← F ∪ {m_i}
    end if
end for
P_r^ref ← ⋃_{m_i∈F} m_i                     ▷Union of foreground masks
return P_r^ref
```

where each mask m_i corresponds to a region in X_i. Naturally, we must determine which mask represents the foreground object in the set $\{m_i\}_{i=1}^k$. For each mask m_i in $\{m_i\}_{i=1}^k$, we first calculate the intersection area between m_i and the initial pseudo-label P_r^{init}. We then calculate the minimum area between m_i and P_r^{init}. Next, we compute the ratio r of the overlap area to the minimum value. The computation process of r can be described as:

$$r = \frac{Area(m_i \cap P_r^{init})}{mim(Area(m_i), Area(P_r^{init}))} , \tag{1}$$

where $Area(\cdot)$ denotes the area of masks, $min(\cdot,\cdot)$ represents calculating the minimum value. If the ratio r exceeds the threshold τ, m_i is classified as a foreground mask. Finally, we take the union of all foreground masks as the optimized RGB pseudo-label P_r^{ref}. The same process can be applied to optimize the thermal pseudo-label P_t^{ref}.

The above process can improve the pseudo-labels to some extent, but it may also introduce redundant information. During the optimization process, due to the characteristics of different modalities and the limitations of SAM, such as the low-light conditions in RGB images or the thermal crossover phenomena in infrared images, SAM may struggle to accurately identify objects. In such cases, when combining initial pseudo-labels for optimization, some regions that do not belong to the objects may be mistakenly retained in the optimized pseudo-labels, as shown in Fig. 3. To reduce irrelevant information, considering the complementary advantages of RGB and thermal images, we first intersect the optimized RGB pseudo-label P_r^{ref} with the initial thermal pseudo-label P_t^{init}. Similarly, we conduct an intersection operation between the optimized thermal pseudo-label P_t^{ref} and the initial RGB pseudo-label P_r^{init}. Then, we merge the results of these intersections through a union operation to acquire the final refined pseudo-label P. This can be described as:

$$P = (P_r^{ref} \cap P_t^{init}) \cup (P_t^{ref} \cap P_r^{init}) . \tag{2}$$

3.3. Progressive Cross-model Cross-scale Fusion Unit (PCCFU)

Most current multi-modal fusion methods insufficiently synergize the interaction between the channel and the spatial dimensions, and they often perform multi-modal feature fusion and cross-scale feature fusion separately. To address these issues, we propose a Progressive Cross-model Cross-scale Fusion Unit (PCCFU). The PCCFU consists of two main components: the Dual Cross-attention Fusion Module (DCFM) and the Cross-Scale Aggregation Module (CSAM). Next, we will introduce each module in detail.

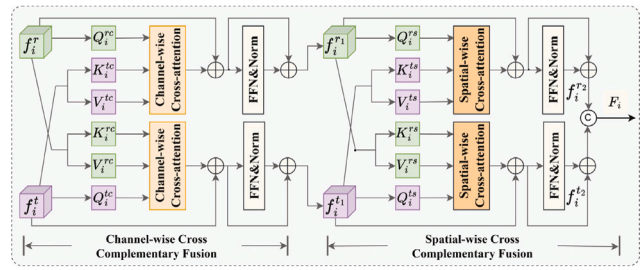

Fig. 4. Dual Cross-attention Fusion Module (DCFM).

3.3.1. Dual Cross-attention Fusion Module

To facilitate the effective collaboration of the two modalities in spatial and channel dimensions, we develop a Dual Cross-attention Fusion Module (DCFM), as shown in Fig. 4. The DCFM employs a cascading structure design. Initially, it utilizes a channel-wise cross-attention mechanism to investigate the correlations between multi-modal features. This process identifies the most relevant channels and effectively suppresses redundant information. Then, based on the previous step, we employ spatial-wise cross-attention between multi-modal features to precisely identify the key areas, effectively avoiding excessive reliance on redundant features.

Unlike the traditional self-attention mechanism in the transformer [25], which has the quadratic time and memory complexity with respect to the input tokens, we employ the transpose attention [47] for cross-modal channel-wise attention and utilize efficient attention [48] for cross-modal spatial-wise cross-attention. These two attention mechanisms are designed to have near-linear memory and computational complexities relative to the number of input tokens. Specifically, given input flattened features $X \in \mathbb{R}^{N\times d}$, where N is the number of tokens and d is the dimensionality of each token. The original self-attention mechanism can be described by the following equation:

$$\begin{aligned} &Q = XW_Q, K = XW_K, V = XW_V, \\ &H = Softmax(\frac{QK^T}{\sqrt{d}})V, \end{aligned} \tag{3}$$

where $W_Q \in \mathbb{R}^{d\times d_q}$, $W_K \in \mathbb{R}^{d\times d_k}$ and $W_V \in \mathbb{R}^{d\times d_v}$ are weight matrices used to linearly project X to query Q, key K and value V, K^T is the transpose of K. The mathematical expression of transpose attention [47] can be described as:

$$XC(Q,K,V) = V * softmax(\frac{K^T Q}{\tau}) , \tag{4}$$

where τ represents the temperature parameter. The mathematical expression of efficient attention [48] can be described as:

$$E(Q,K,V) = \rho_q(Q)\left(\rho_k(K)^T V\right) , \tag{5}$$

where $\rho_q(\cdot)$ and $\rho_k(\cdot)$ represent the normalization functions for the query Q and key K, respectively. For further details about the above attention mechanisms, readers are referred to the relevant literature [25,47,48].

Within the DCFM, for the input multi-modal features $f_i^r \in \mathbb{R}^{H\times W\times C}$ and $f_i^t \in \mathbb{R}^{H\times W\times C}$, we first flatten them with size $\mathbb{R}^{H\times W\times C}$ to $\mathbb{R}^{N\times C}$, where $N = H\times W$. Then we carry out the channel-wise cross-attention interaction between f_i^r and f_i^t, obtaining $f_i^{r_1}$ and $f_i^{t_1}$. The mathematical expression for $f_i^{r_1}$ is as follows:

$$\begin{aligned} &f_i^{rc} = XC(Q_i^{rc}, K_i^{tc}, V_i^{tc}) + f_i^r, \\ &f_i^{r_1} = FFN(Norm(f_i^{rc})) + f_i^{rc}, \end{aligned} \tag{6}$$

where $XC(\cdot)$ means the transpose attention, '+' means the residual connection, $Norm(\cdot)$ means the normalization layers, $FFN(\cdot)$ means the feed forward neural layer. The calculation formula for $f_i^{t_1}$ is similar to

$f_i^{r_1}$. Next, we perform spatial-wise cross-attention between $f_i^{r_1}$ and $f_i^{t_1}$, obtaining $f_i^{r_2}$ and $f_i^{t_2}$. The formulation of $f_i^{r_2}$ is as follows:

$$\begin{aligned} f_i^{rs} &= E(Q_i^{rs}, K_i^{ts}, V_i^{ts}) + f_i^{r_1} \ , \\ f_i^{r_2} &= FFN(Norm(f_i^{rs})) + f_i^{rs} \ , \end{aligned} \tag{7}$$

where $E(\cdot)$ denotes the efficient attention. The calculation formula for $f_i^{t_2}$ is similar to $f_i^{r_2}$.

Finally, we concatenate the two features $f_i^{r_2}$ and $f_i^{t_2}$ along the channel dimension and we reduce the channel dimension to 128 to lower computational costs. The final fusion features F_i can be formulated as:

$$F_i = ReLU(BN(Conv_1(Conv_3([f_i^{r_2}, f_i^{t_2}])))), \tag{8}$$

where $[\cdot]$ is the concatenate operation, $Conv_1(\cdot)$ is the 1×1 convolution, $Conv_3(\cdot)$ is the 3×3 convolution, $BN(\cdot)$ is the batch normalization and $ReLU(\cdot)$ is the ReLU activation function.

3.3.2. Cross-Scale Aggregation Module

Features from different levels exhibit distinct characteristics. High-level features provide global contextual information, while low-level features offer fine details. Most existing methods neglect cross-level feature interaction before decoding, therefore, we propose a Cross-scale Aggregation Module (CSAM) to stepwise integrate global and local information using cross-attention operation before decoding.

As shown in Fig. 5, we derive mixed features F_i, F_{i+1}, $\cdots$, F_4 from the DCFM of each PPCFU. For the CSAM in the ith PCCFU, if there exists the output feature $\hat{F}_{i+1}$ from the $(i+1)^{th}$ PCCFU, the output $\hat{F}_i$ of the ith PCCFU is obtained through cross-scale interaction between F_i and the outputs of all previous PCCFUs. Otherwise, the output $\hat{F}_i$ is equal to F_i. This process is summarized as follows:

$$\hat{F}_i = \begin{cases} CSAM(F_4) = F_4, & i = 4 \ , \\ CSAM\left(F_i, \hat{F}_{i+1}, \ldots, \hat{F}_4\right), & i = 1, 2, 3. \end{cases} \tag{9}$$

When conducting cross-scale fusion in CSAM, our cross-attention operation is also implemented by efficient attention mechanism [48]. For the CSAM in the ith PCCFU, the output $\hat{F}_i$ is obtained through cross-attention operation among features F_i, $\hat{F}_{i+1}$, $\ldots$, $\hat{F}_4$. Specifically, for feature $F_i (i \in 1, 2, 3)$, we linearly project it to $Q_{i+1}, \ldots, Q_4$ through different weight metrics $W_Q^{i+1}, \ldots, W_Q^4$; for feature $F_j (j \in i+1, \ldots, 4)$, we linearly project it to K_j and V_j, using two weight matrices W_K^j and W_V^j. This can be denoted as:

$$\begin{aligned} &Q_{i+1} = F_i W_Q^{i+1}, \ldots, Q_4 = F_i W_Q^4 \ , \\ &K_j = F_j W_K^j, V_j = F_j W_V^j \ , \ j = i+1, \ldots, 4 \ . \end{aligned} \tag{10}$$

Subsequently, cross-attention is performed on Q_{i+1}, K_{i+1}, V_{i+1}; $\ldots$; and Q_4, K_4, V_4, respectively. The results are then concatenated, with a residual connection to F_i. Finally, the output is normalized and passed through a feed forward neural layer. The output $\hat{F}_i$ of the ith PCCFU can be expressed as follows:

$$\begin{aligned} &\bar{F}_j = E(Q_j, K_j, V_j), \ j = i+1, \ldots, 4 \ , \\ &F_i' = BN(ReLU([\bar{F}_{i+1}, \ldots, \bar{F}_4])) \ , \\ &F_i' = Conv_1(Conv_3(F_i')) + F_i, \\ &\hat{F}_i = FFN(Norm(F_i')) + F_i' \ . \end{aligned} \tag{11}$$

3.4. Decoder and loss function

After obtaining the fused features $\hat{F}_i$ $(i \in \{1, 2, 3, 4\})$ from the PPCFUs, we start from the deepest feature map $\hat{F}_4$, progressively upsampling and fusing with shallower features through the concatenation operation to generate the final saliency map P. This process is formulated as follows:

$$\tilde{F}_i = \begin{cases} Conv_3(Up_2(\hat{F}_i)), & i = 4 \ , \\ Conv_1(Conv_3(Up_2([\hat{F}_i, \tilde{F}_{i+1}]))), & i = 1, 2, 3. \end{cases} \tag{12}$$

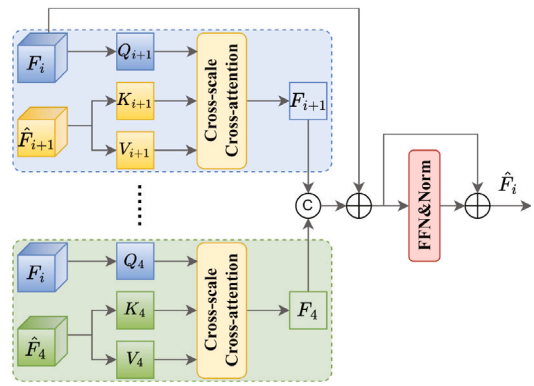

Fig. 5. Cross-scale Aggregation Module (CSAM).

$$P = Conv_1(Up_4(\tilde{F}_1)), \tag{13}$$

where $Up_x(\cdot)$ represents upsampling by a factor of x.

During the training process, we employ the following defined loss function to train our model:

$$L = L_{ppa}(P, Y) + L_{pce}(P, S), \tag{14}$$

where P is the predicted result, Y is the pseudo-label generated by SGLO, S is the scribble annotation. The pixel position aware loss [49] can be formulated as:

$$L_{ppa}(P, Y) = L_{wbce}(P, Y) + L_{wiou}(P, Y), \tag{15}$$

where L_{wbce} is the weighted binary cross entropy loss, L_{wiou} is the weighted IoU loss. The partial cross-entropy loss [50] can be formulated as:

$$L_{pce}(P, S) = \sum_{(u,v) \in J_l} L_{(u,v)}(P, S) \ , \tag{16}$$

where J_l denotes the labeled pixels, (u, v) represents pixel coordinates, and $L_{(u,v)}$ is the cross-entropy loss at (u, v).

4. Experimental results

4.1. Experiment settings

4.1.1. Datasets and evaluation metrics

We utilize the RGBT-S dataset, annotated by Liu et al. [11], as our training dataset. This dataset is derived from the reannotation of the VT5000 benchmark training set [51] using scribble annotations. To assess the performance of various methods, we conduct evaluations on three RGBT SOD testing datasets: VT821 [52], VT1000 [53], and the testing dataset of VT5000. Furthermore, to prove the effectiveness of our approach, we also conduct the experiments on the scribble-supervised RGBD SOD task. For this task, we employ the RGBD scribble datasets NJU2K-S and NLPR-S provided by Xu et al. [17] as training datasets. We then evaluate the methods on six RGBD benchmarks, namely DES [54], NLPR [55], SIP [56], STERE [57], SSD [58] and LFSD [59].

To assess the performance of various methods, we employ widely used SOD metrics, including the S-measure (S_m) [60], mean F-measure(F_β^{avg}) [61], mean E-measure (E_m^{avg}) [62], weighted F-measureF_β^ω [63], and mean absolute error (MAE) [64]. Higher values of S_m, F_β^{avg}, E_m^{avg} and F_β^ω indicate the superior prediction capability of the model. Conversely, a smaller MAE value signifies better performance. Additionally, we plot Precision–Recall (PR) curves and F_β-Threshold curves [65] across various thresholds. These curves aid in analyzing the comprehensive performance of the model with different thresholds.

Table 1
Quantitative comparison with recent state-of-the-art fully and weakly supervised RGBT SOD methods. The best results are represented in **bold**, and the second-best results are indicated with underlining.

Methods	VT821					VT1000					VT5000-Test				
	$S_m \uparrow$	$F_\beta^\omega \uparrow$	$MAE \downarrow$	$F_\beta^{avg} \uparrow$	$E_m^{avg} \uparrow$	$S_m \uparrow$	$F_\beta^\omega \uparrow$	$MAE \downarrow$	$F_\beta^{avg} \uparrow$	$E_m^{avg} \uparrow$	$S_m \uparrow$	$F_\beta^\omega \uparrow$	$MAE \downarrow$	$F_\beta^{avg} \uparrow$	$E_m^{avg} \uparrow$
Fully supervised RGBT SOD methods															
$SwinNet_{22}$ [9]	0.904	0.818	0.030	0.861	0.926	0.938	0.894	0.018	0.916	0.965	0.912	0.846	0.026	0.882	0.944
$CCFENet_{22}$ [13]	0.900	0.852	0.027	0.865	0.932	0.934	0.910	0.018	0.913	0.964	0.896	0.849	0.030	0.867	0.935
$HRTrans_{23}$ [10]	0.906	0.849	0.026	0.866	0.935	0.938	0.913	0.017	0.915	0.967	0.912	0.870	0.025	0.883	0.950
$CAVER_{23}$ [15]	0.898	0.846	0.026	0.861	0.930	0.938	0.912	0.016	0.916	0.967	0.899	0.849	0.028	0.866	0.939
$SPNet_{23}$ [40]	0.913	0.873	0.023	0.881	0.944	0.941	0.925	0.015	0.924	0.972	0.914	0.880	0.024	0.892	0.951
$WaveNet_{23}$ [43]	0.912	0.861	0.025	0.871	0.937	0.945	0.920	0.015	0.922	0.970	0.911	0.862	0.026	0.879	0.944
$SACNet_{24}$ [44]	0.903	0.856	0.025	0.864	0.935	0.941	0.923	0.014	0.923	0.972	0.916	0.884	**0.021**	0.895	0.956
$WGOFNet_{24}$ [42]	0.908	0.868	0.025	0.880	0.940	0.940	0.922	0.016	0.923	0.969	0.911	0.873	0.025	0.888	0.947
$MCNet_{24}$ [41]	**0.923**	**0.881**	**0.021**	0.890	**0.948**	**0.948**	**0.929**	**0.013**	0.930	**0.975**	**0.924**	**0.891**	**0.021**	**0.902**	**0.957**
Weakly supervised RGBT SOD methods															
$WSSA_{20}$ [16]	0.816	0.692	0.066	0.730	0.851	0.885	0.812	0.037	0.838	0.917	0.809	0.674	0.064	0.716	0.852
$DENet_{22}$ [17]	0.810	0.668	0.074	0.710	0.829	0.893	0.815	0.034	0.849	0.923	0.827	0.693	0.057	0.743	0.867
$MIRV_{23}$ [35]	0.841	0.749	0.043	0.778	0.870	0.905	0.867	0.026	0.882	0.943	0.842	0.758	0.047	0.787	0.893
$LEPAS_{23}$ [11]	0.895	0.855	0.027	0.873	0.935	0.925	0.905	0.020	0.915	0.960	0.877	0.829	0.033	0.855	0.928
Ours	0.912	0.876	0.024	**0.896**	0.946	0.943	**0.929**	0.016	**0.938**	0.972	0.907	0.874	0.026	0.898	0.948

4.1.2. Implementation details

We employ the Shunted Transformer [45], pretrained on ImageNet [66], as our encoder to extract features from different modalities. All experiments are conducted using PyTorch, based on Python 3.10, and the training process is performed on a single NVIDIA GeForce RTX 3090 GPU. The input image size is set to 256 × 256 for both training and testing phases. Our network undergoes training for 100 epochs using the Adam optimizer [67] with a batch size of 12. The initial learning rate is set to 5e–5, and is scheduled to be decreased by a factor of 10 every 40 epochs. Additionally, in the second optimization stage of SGLO, we set the threshold τ to 0.85 for selecting the foreground mask from the SAM's automatic segmentation results.

4.2. Quantitative comparison with others

To evaluate the performance on the RGBT SOD task, we conduct a comprehensive comparison between our proposed method and thirteen recent approaches. Nine of them are fully supervised RGBT SOD methods, including SwinNet [9], HRTrans [10], CCFENet [13], CAVER [15], SPNet [40],WaveNet [43], SACNet [44], MCNet [41] and WGOFNet [42]. Additionally, four scribble-supervised SOD methods are included: LEPAS [11], WSSA [16], DENet [17], and MIRV [35]. For RGB-based SOD method WSSA, we fuse RGBT image pairs as model's inputs and use the provided source code to retrain the model. For DENet and MIRV, originally designed for RGBD SOD task, we replace the depth modality with thermal modality and retrain these methods on the RGBT-S [11] dataset using the provided code.

As illustrated in Table 1, three key observations can be drawn from the results. First, there is a notable performance gap between weakly supervised methods and fully supervised ones, with the latter achieving significantly better results. For instance, the state-of-the-art fully supervised method MCNet [41] outperforms the best weakly supervised method LEPAS [11] by average improvements of 3.7% (S_m), 4.4% (F_β^w), 3.0% (F_β^{avg}), 46.5% (MAE) and 2.0% (E_m^{avg}) across three datasets. Second, our method significantly surpasses the current SOTA weakly supervised method LEPAS [11], with an average improvements of 2.4%(S_m), 3.5% (F_β^w), 3.4% (F_β^{avg}), 21.5% (MAE) and 1.5% (E_m^{avg}) across three datasets. Third, our method also outperforms most fully supervised RGBT SOD methods. On the VT5000 dataset, our method is only 0.4% lower than the current SOTA method MCNet [41] on the F_β^{avg} metric, and demonstrates greater superiority on the VT821 and VT1000 datasets. Moreover, Fig. 6 presents the Precision–Recall (PR) and F_β-Threshold curves for various methods. The positioning of these curves indicates that, despite employing a weakly supervised approach, our performance keeps robust compared to the other methods evaluated.

To validate the generalizability of our proposed weakly supervised approach, we conduct a comparative analysis against six state-of-the-art scribble-supervised RGBD SOD methods, namely DENet [17], LEPAS [11], MIRV [35], DHFR [68], LGR [69] and RPPS [70]. Due to the unavailability of source codes of some methods, we rely on the experimental results presented in their respective papers. A detailed quantitative comparison in Table 2 demonstrates that our method achieves superior performance across various evaluation metrics and datasets. Although our method may slightly underperforms on certain metrics for specific datasets, its overall performance is the best among the compared methods. For instance, considering the F_β^{avg} metric, we compute the average F_β^{avg} scores across all datasets for each method. Our method outperforms the current SOTA method LGR [69] by 1.6%, demonstrating a significant improvement on the average F_β^{avg} score compared with other scribble-supervised RGBD SOD methods.

Tables 1 and 2 clearly illustrate that our method has attained SOTA performance in both scribble-supervised RGBT and RGBD SOD methods. Moreover, it exhibits competitive results compared to fully supervised methods, thereby affirming the efficacy of our proposed pseudo-label optimization and multi-modal fusion strategies.

4.3. Challenges-based quantitative comparison

The VT5000 dataset includes eleven challenging scenarios. According to the description in [71], the specific challenges include big salient objects (BSO), small salient objects (SSO), multiple salient objects (MSO), low illumination environments (LI), center bias (CB), crossing image boundaries (CIB), similar appearances (SA), thermal crossover (TC), image clutter (IC), out-of-focus areas (OF), and adverse weather conditions (BW). In addition, the VT5000 dataset includes annotations for low-quality RGB images (BR) and low-quality thermal images (BT).

To assess the robustness of our method, we present the evaluation results of various methods across challenging scenarios in Table 3. We utilize the mean F-measure (F_β^{avg}) as our evaluation metric to reflect the overall performance. As shown in Table 3, it is clear that in nearly all challenging scenarios, both our approach and the current fully supervised methods significantly outperform the SOTA weakly supervised method LEPAS [11]. Furthermore, our method ranks second only to the SOTA fully supervised method, MCNet [41], and even surpasses MCNet in certain specific scenarios (such as LI, SA, and OF). This demonstrates that the pseudo-labels produced by our SGLO are comparable to actual pixel-wise annotations, and the fused features derived from our designed PCCFU exhibit exceptional robustness.

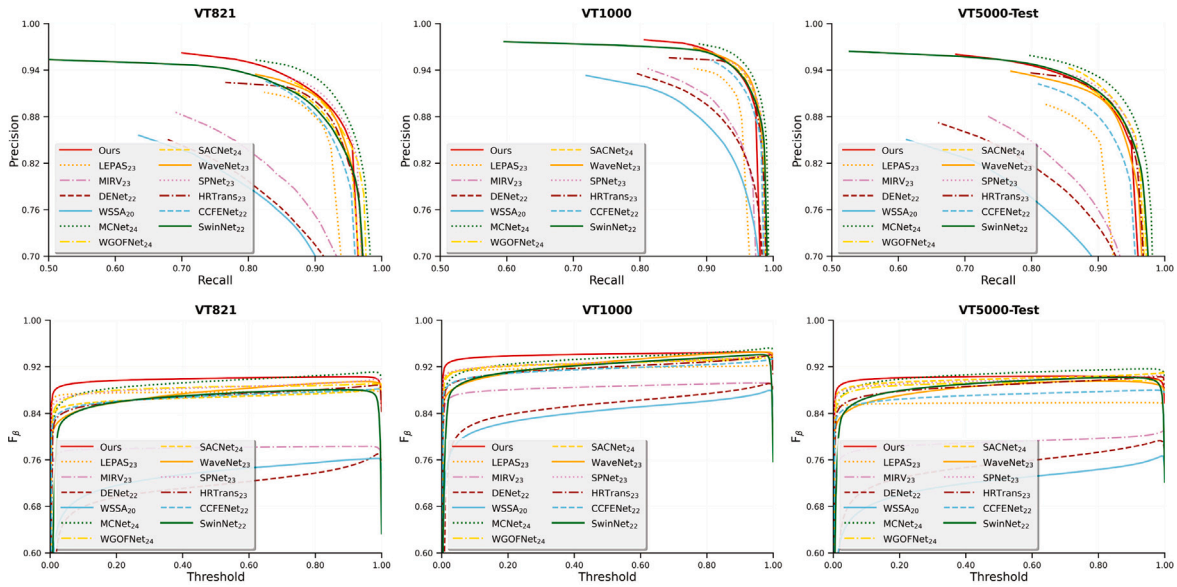

Fig. 6. The Precision–Recall(PR) and F_β-Threshold curves of different methods.

Table 2
Quantitative comparison with the recent state-of-the-art weakly supervised RGBD SOD methods. The best results are highlighted in **bold**, and the second-best results are indicated with underlining.

Methods	DES				NLPR				SIP				STERE				SSD				LFSD			
	$S_m \uparrow$	$F_\beta^{avg} \uparrow$	$E_m^{avg} \uparrow$	$MAE \downarrow$	$S_m \uparrow$	$F_\beta^{avg} \uparrow$	$E_m^{avg} \uparrow$	$MAE \downarrow$	$S_m \uparrow$	$F_\beta^{avg} \uparrow$	$E_m^{avg} \uparrow$	$MAE \downarrow$	$S_m \uparrow$	$F_\beta^{avg} \uparrow$	$E_m^{avg} \uparrow$	$MAE \downarrow$	$S_m \uparrow$	$F_\beta^{avg} \uparrow$	$E_m^{avg} \uparrow$	$MAE \downarrow$	$S_m \uparrow$	$F_\beta^{avg} \uparrow$	$E_m^{avg} \uparrow$	$MAE \downarrow$
$DENet_{22}$ [17]	.916	.887	.939	.029	.913	.874	.936	.028	.865	.841	.900	.056	.887	.851	.914	.044	.850	.799	.873	.061	.832	.814	.860	.085
$LEPAS_{23}$ [11]	.887	.875	.937	.029	.901	.890	.953	.027	.820	.827	.902	.072	.885	.893	.940	.044	.834	.815	.904	.063	.861	.868	.906	.065
$MIRV_{23}$ [35]	.928	.921	.965	.019	.914	.895	.953	.025	.876	.863	.924	.049	.891	.874	.937	.042	–	–	–	–	.849	.844	.889	.072
$DHFR_{23}$ [68]	.915	.930	.956	.022	.904	.901	.950	.027	.843	.874	.902	.064	.884	.896	.935	.043	.858	**.869**	.911	.051	.855	.878	.904	.066
$RPPS_{23}$ [70]	–	–	–	–	.899	.884	.947	.035	.876	.872	.925	.060	.881	.870	.929	.059	.864	.849	.918	.061	.835	.837	.879	.095
LGR_{24} [69]	.923	.912	.965	.020	**.920**	.894	**.959**	**.023**	.855	.881	.933	.044	**.901**	.895	.942	**.038**	–	–	–	–	.862	.869	.905	.062
Ours	**.933**	**.940**	**.979**	**.017**	.915	**.914**	.956	.026	**.901**	**.923**	**.951**	**.037**	.896	**.904**	**.946**	.041	**.872**	.858	**.930**	**.044**	**.875**	**.888**	**.913**	**.061**

Table 3
Quantitative comparison with the recent state-of-the-art methods on challenging scenarios. The best results are highlighted in **bold**, and the second-best results are indicated with underlining. The evaluation metric is mean F-measure F_β^{avg}.

Methods	BSO	BW	CB	CIB	IC	LI	MSO	OF	SA	SSO	TC	BR	BT
Fully supervised RGBT SOD methods													
$SwinNet_{22}$ [9]	0.904	0.831	0.875	0.895	0.857	0.898	0.860	0.872	0.867	0.812	0.867	0.883	0.886
$CCFENet_{22}$ [13]	0.887	0.802	0.859	0.873	0.830	0.875	0.837	0.859	0.859	0.820	0.854	0.870	0.870
$HRTrans_{23}$ [10]	0.901	0.816	0.881	0.889	0.857	0.878	0.860	0.858	0.873	0.838	0.873	0.886	0.886
$CAVER_{23}$ [15]	0.892	0.834	0.863	0.885	0.833	0.881	0.841	0.846	0.840	0.792	0.847	0.870	0.869
$SPNet_{23}$ [40]	0.910	0.843	0.891	0.902	0.863	**0.904**	0.874	0.884	0.882	0.845	0.883	0.894	0.895
$WaveNet_{23}$ [43]	0.901	0.837	0.881	0.892	0.852	0.893	0.864	0.866	0.869	0.824	0.861	0.882	0.884
$SACNet_{24}$ [44]	0.913	0.848	0.889	0.907	0.875	0.903	0.877	0.879	0.883	0.848	0.887	0.897	0.898
$WGOFNet_{24}$ [42]	0.908	0.816	0.882	0.898	0.857	0.896	0.866	0.881	0.870	0.853	0.868	0.891	0.892
$MCNet_{24}$ [41]	**0.918**	**0.865**	**0.901**	**0.912**	**0.883**	0.903	**0.887**	0.881	0.879	0.855	**0.893**	**0.905**	**0.905**
Weakly supervised RGBT SOD methods													
$WSSA_{20}$ [16]	0.785	0.642	0.693	0.743	0.667	0.760	0.694	0.725	0.705	0.537	0.687	0.717	0.723
$DENet_{22}$ [17]	0.808	0.677	0.729	0.781	0.704	0.780	0.730	0.746	0.729	0.545	0.708	0.744	0.750
$MIRVN_{23}$ [35]	0.822	0.715	0.788	0.814	0.747	0.786	0.774	0.772	0.769	0.689	0.767	0.794	0.792
$LEPAS_{23}$ [11]	0.873	0.775	0.857	0.870	0.828	0.826	0.834	0.819	0.807	0.811	0.847	0.863	0.857
Ours	0.913	0.847	0.896	0.907	0.870	0.899	0.873	**0.898**	**0.887**	**0.864**	0.887	0.899	0.901

4.4. Qualitative comparison with others

We select 10 samples that present challenging scenarios and display the prediction results of various methods in Fig. 7. These challenging scenarios include: crossing image boundaries(*rows* (a) and (b)), center bias(*rows* (c) and (d)), thermal crossover(*rows* (a), (e), (g) and (i)), similar appearance(*row* (g)), shadows in RGB images(*row* (h)), big salient objects(*rows* (a), (f) and (j)) and multiple salient objects(*row* (j)).

As shown in Fig. 7, our method demonstrates superior adaptability in accurately segmenting salient objects across various challenging scenarios, outperforming competing techniques. Specifically, as shown in the *row* (j) of Fig. 7, this example contains challenges such as large, multiple and small objects. Our approach can detect complete object and preserve fine details, as well as effectively capturing intricate features. In contrast, methods such as MCNet [41], SPNet [40], WaveNet [43] and LEPAS [11] exhibit notable deficiencies in handling such complex situations. Furthermore, *rows* (e), (g) and (i) show

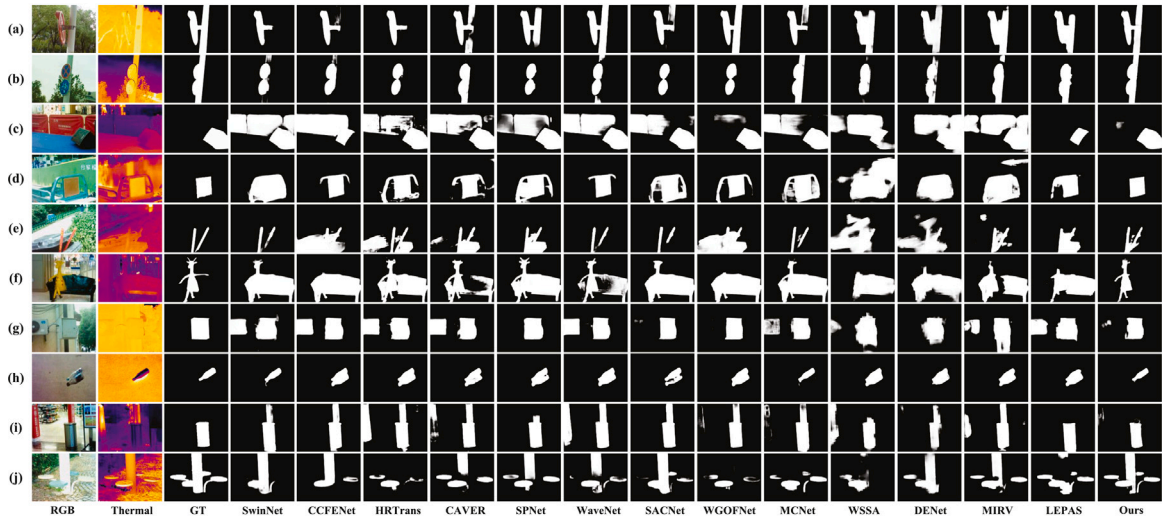

Fig. 7. Qualitative comparison of different methods on challenging scenarios.

Table 4
Ablation analysis of the proposed SGLO. The initial labels and final labels are generated from different stages of SGLO. The best results are highlighted in **bold**.

Supervision labels	VT821		VT1000		VT5000-Test	
	S_m ↑	F_β^{avg} ↑	S_m ↑	F_β^{avg} ↑	S_m ↑	F_β^{avg} ↑
Scribbles	0.862	0.802	0.897	0.870	0.848	0.793
+ initial labels	0.897	0.874	0.928	0.925	0.888	0.882
+ final labels	**0.912**	**0.896**	**0.943**	**0.938**	**0.907**	**0.898**

Table 5
Ablation analysis for the threshold τ of selecting the foreground masks in the second optimization stage of SGLO. The experiments are conducted on VT821 dataset and the best results are highlighted in **bold**.

Threshold	S_m ↑	F_β^{avg} ↑
$\tau = 0.80$	0.9012 ± 0.0011	0.8873 ± 0.0009
$\tau = 0.81$	0.8994 ± 0.0015	0.8892 ± 0.0013
$\tau = 0.82$	0.9047 ± 0.0010	0.8905 ± 0.0007
$\tau = 0.83$	0.9075 ± 0.0008	0.8929 ± 0.0009
$\tau = 0.84$	0.9061 ± 0.0013	0.8913 ± 0.0011
$\tau = 0.85$	**0.9114 ± 0.0010**	**0.8948 ± 0.0008**
$\tau = 0.86$	0.9087 ± 0.0014	0.8923 ± 0.0011
$\tau = 0.87$	0.9051 ± 0.0013	0.8918 ± 0.0010
$\tau = 0.88$	0.9065 ± 0.0009	0.8876 ± 0.0012
$\tau = 0.89$	0.9083 ± 0.0012	0.8922 ± 0.0009
$\tau = 0.90$	0.9053 ± 0.0011	0.8931 ± 0.0008

the challenges such as thermal crossover and similar appearances, which often lead to inconsistent saliency detection across RGB and thermal modalities. Our method leverages the information from both modalities for more accurate identification of salient objects, whereas other techniques are influenced by noises, resulting in poor predictions. Additionally, in rows (c) and (h), other methods frequently detect the false salient object, while our approach does not present such problem. Overall, although our method may face minor noise interference (*rows* (c), (e), (g), and (i)) or slight information loss (*row*(j)), it consistently achieves superior accuracy and robustness in detection compared with other methods.

4.5. Ablation studies

4.5.1. Effectiveness of the SGLO

To validate the efficacy of our proposed two-stage SAM-Guided Label Optimization strategy (SGLO), we utilize various types of labels for training. The first type uses only scribble annotations. The second type incorporates scribble annotations and initial labels, which are the intersection of the initial RGB and thermal pseudo-labels obtained from the first stage of SGLO. Comparing the 1st and 2nd rows in Table 4, the incorporation of initial labels significantly improved the model's performance. The third type employs scribble annotations and final labels, which are the pseudo-labels generated by SGLO. By comparing the 2nd and 3rd rows, it is evident that the performance has further improved. The above analysis demonstrates that the two stages of our proposed SGLO are both effective. In addition, we discuss the threshold τ for selecting the foreground mask in the second optimization stage of SGLO. As illustrated in Table 5, we compare the method's performance on the VT821 dataset across different thresholds and find that the method achieves the best result when the threshold τ is set to 0.85.

To provide a deeper comparison of these methods, we add confidence intervals to the data presented in Tables 5, 7, 8, following the methodologies in [17,72]. Each method undergoes five independent tests, and we compute the 95% confidence intervals, retaining four decimal places.

Table 6
Ablation analysis of the contribution of different components in our model. The best results are highlighted in bold. w/o denotes the removal of the module from the full model.

Models	VT821		VT1000		VT5000-Test	
	S_m ↑	F_β^{avg} ↑	S_m ↑	F_β^{avg} ↑	S_m ↑	F_β^{avg} ↑
Ours	**0.912**	**0.896**	**0.943**	**0.938**	**0.907**	**0.898**
w/o DCFM	0.904	0.885	0.938	0.930	0.898	0.889
w/o CSAM	0.902	0.881	0.934	0.931	0.895	0.887

4.5.2. Effectiveness of the DCFM

In this section, we employ the scribbles and the pseudo-labels derived from SGLO as supervisory information. To ascertain the effectiveness of the DCFM, we exclude it from our full model (labeled as w/o DCFM) and utilize concatenation to integrate multi-modal features. By comparing the 1st and 2nd rows in Table 6, we observe that the removal of DCFM results in a degradation of the performance by 1.2%, 0.9%, and 1.0% in terms of F_β across the three datasets.

Furthermore, to validate the rationality of the structure design of DCFM, we have developed four variations: using spatial-wise cross-attention only, using channel-wise cross-attention only, executing spatial-wise and channel-wise cross-attention interactions in parallel, and first performing spatial-wise and then channel-wise cross-attention interactions. As indicated in Table 7, the combination of channel-wise and spatial-wise cross-attention surpasses the performance of methods

Table 7
Ablation analysis on the rationality of DCFM design. The best results are highlighted in **bold**. 'C' denotes channel-wise cross-attention, while 'S' indicates spatial-wise cross-attention.

Models	VT821		VT1000		VT5000-Test	
	S_m ↑	F_β^{avg} ↑	S_m ↑	F_β^{avg} ↑	S_m ↑	F_β^{avg} ↑
C	0.9055 ± 0.0011	0.8903 ± 0.0008	0.9398 ± 0.0007	0.9328 ± 0.0005	0.8999 ± 0.0006	0.8918 ± 0.0005
S	0.9072 ± 0.0012	0.8866 ± 0.0009	0.9386 ± 0.0007	0.9321 ± 0.0006	0.8981 ± 0.0006	0.8932 ± 0.0007
S+C(Cascade)	0.9075 ± 0.0008	0.8902 ± 0.0010	0.9409 ± 0.0005	0.9343 ± 0.0004	0.9039 ± 0.0004	0.8940 ± 0.0005
S+C(Parallel)	0.9079 ± 0.0016	0.8921 ± 0.0013	0.9389 ± 0.0008	0.9358 ± 0.0007	0.9021 ± 0.0007	0.8949 ± 0.0006
C+S(Cascade)	**0.9114 ± 0.0010**	**0.8948 ± 0.0008**	**0.9427 ± 0.0006**	**0.9373 ± 0.0004**	**0.9067 ± 0.0005**	**0.8981 ± 0.0003**

Table 8
Ablation studies on cross-scale feature fusion methods. The best results are highlighted in **bold**.

Models	VT821		VT1000		VT5000-Test	
	S_m ↑	F_β^{avg} ↑	S_m ↑	F_β^{avg} ↑	S_m ↑	F_β^{avg} ↑
Concatenation	0.9045 ± 0.0015	0.8824 ± 0.0012	0.9374 ± 0.0009	0.9315 ± 0.0006	0.9022 ± 0.0007	0.8934 ± 0.0005
Addition	0.9080 ± 0.0011	0.8903 ± 0.0009	0.9402 ± 0.0005	0.9329 ± 0.0004	0.9018 ± 0.0004	0.8912 ± 0.0005
Cross-attention	**0.9114 ± 0.0010**	**0.8948 ± 0.0008**	**0.9427 ± 0.0006**	**0.9373 ± 0.0004**	**0.9067 ± 0.0005**	**0.8981 ± 0.0003**

utilizing either alone. By comparing the 3rd and 5th rows, we observe that the cascaded structural design outperforms the parallel structural design.

Moreover, comparing the 4th and 5th rows demonstrates that the application of channel-wise cross-attention fusion prior to spatial-wise cross-attention fusion yields superior performance.

4.5.3. Effectiveness of the CSAM

To verify the CSAM is effective, we remove the CSAM from our full model(w/o CSAM). Comparing the 1st and 3rd rows in Table 6, we find that removing the CSAM decreases the performance of the model by 1.7%, 0.7% and 1.3% on F_β^{avg} across three datasets. In our designed CSAM, we employ the cross-attention mechanism to merge features from both lower and higher levels. To validate the efficacy of this cross-attention mechanism, we contrast it with two alternative operations: concatenation and addition. As indicated in Table 8, the cross-attention mechanism yields superior performance when compared to the direct use of concatenation or addition operations.

5. Discussion for limitations and future work

Although our approach has yielded very promising results, the main limitation is the reliance on a widely adopted two-stage training strategy. In this framework, the pseudo-labels produced in the first stage are crucial for the effectiveness of the subsequent training phase. However, it is unable to dynamically optimize the quality of these pseudo-labels during the second stage of model learning. In the future work, we aim to develop a one-stage end-to-end method that incorporates an adaptive prompts generation module within the network. This module will automatically create prompts tailored for SAM based on image features and utilize multi-level features to guide the segmentation process. By employing an end-to-end joint training, the prompts generation and SAM's outputs will mutually optimize each other.

6. Conclusion

In this work, we explore the weakly supervised RGBT salient object detection using scribble information. To generate pseudo-labels close to the manually annotated ground truth, we propose a Two-stage SAM-guided Label Optimization (SGLO) strategy. And we design a method to refine the initial pseudo-labels with multi-modality information in the second stage of SGLO. To fully synergize the interaction of multi-modal features across different dimensions and progressively integrate all higher-level features during the fusion process, we propose a Progressive Cross-modal Cross-scale Fusion Unit which consists of a Dual Cross-attention Fusion Module and a Cross-scale Aggregation Module. The former facilitates the collaborative fusion from both channel and spatial dimensions, while the latter enables lower-level features to interact with all higher-level features during the multi-modal fusion. Compared with recent weakly supervised RGBT or RGBD methods, our method not only achieves a meaningful improvement over them but also outperforms the majority of fully supervised methods.

CRediT authorship contribution statement

Sulan Zhai: Supervision, Conceptualization. **Chengzhuang Liu:** Writing – original draft, Methodology. **Zhengzheng Tu:** Writing – review & editing, Funding acquisition. **Chenglong Li:** Validation, Formal analysis. **Liuxuanqi Gao:** Visualization, Investigation.

Declaration of competing interest

All authors disclosed no relevant relationships.

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Data availability

The data that has been used is confidential.

References

[1] L. Guo, P. Shi, L. Chen, C. Chen, W. Ding, Pixel and region level information fusion in membership regularized fuzzy clustering for image segmentation, Inf. Fusion 92 (2023) 479–497, http://dx.doi.org/10.1016/j.inffus.2022.12.008.
[2] S. Gupta, Y. Zhang, X. Hu, P. Prasanna, C. Chen, Topology-aware uncertainty for image segmentation, in: Adv. Neural Inf. Process. Syst., 2023, pp. 8186–8207.
[3] R.P. Babadian, K. Faez, M. Amiri, E. Falotico, Fusion of tactile and visual information in deep learning models for object recognition, Inf. Fusion 92 (2023) 313–325, http://dx.doi.org/10.1016/j.inffus.2022.11.032.
[4] D. Li, J. Hu, C. Wang, X. Li, Q. She, L. Zhu, T. Zhang, Q. Chen, Involution: Inverting the inherence of convolution for visual recognition, in: IEEE Conf. Comput. Vis. Pattern Recognit., 2021, pp. 12316–12325, http://dx.doi.org/10.1109/CVPR46437.2021.01214.
[5] P. Sun, H. Kretzschmar, X. Dotiwalla, A. Chouard, V. Patnaik, P. Tsui, J. Guo, Y. Zhou, Y. Chai, B. Caine, et al., Scalability in perception for autonomous driving: Waymo open dataset, in: IEEE Conf. Comput. Vis. Pattern Recognit., 2020, pp. 2443–2451, http://dx.doi.org/10.1109/CVPR42600.2020.00252.
[6] H. Caesar, V. Bankiti, A.H. Lang, S. Vora, V.E. Liong, Q. Xu, A. Krishnan, Y. Pan, G. Baldan, O. Beijbom, Nuscenes: A multimodal dataset for autonomous driving, in: IEEE Conf. Comput. Vis. Pattern Recognit., 2020, pp. 11618–11628, http://dx.doi.org/10.1109/CVPR42600.2020.01164.

[7] X. Wang, S. Li, C. Chen, Y. Fang, A. Hao, H. Qin, Data-level recombination and lightweight fusion scheme for RGB-D salient object detection, IEEE Trans. Image Process. 30 (2020) 458–471, http://dx.doi.org/10.1109/TIP.2020.3037470.

[8] J. Zhang, D.-P. Fan, Y. Dai, S. Anwar, F. Saleh, S. Aliakbarian, N. Barnes, Uncertainty inspired RGB-D saliency detection, IEEE Trans. Pattern Anal. Mach. Intell. 44 (2021) 5761–5779, http://dx.doi.org/10.1109/TPAMI.2021.3073564.

[9] Z. Liu, Y. Tan, Q. He, Y. Xiao, SwinNet: Swin transformer drives edge-aware RGB-D and RGB-T salient object detection, IEEE Trans. Circuits Syst. Video Technol. 32 (2022) 4486–4497, http://dx.doi.org/10.1109/TCSVT.2021.3127149.

[10] B. Tang, Z. Liu, Y. Tan, Q. He, HRTransNet: Hrformer-driven two-modality salient object detection, IEEE Trans. Circuits Syst. Video Technol. 33 (2023) 728–742, http://dx.doi.org/10.1109/TCSVT.2022.3202563.

[11] Z. Liu, X. Huang, G. Zhang, X. Fang, L. Wang, B. Tang, Scribble-supervised RGB-T salient object detection, in: IEEE Int. Conf. Multimedia Expo., 2023, pp. 2369–2374, http://dx.doi.org/10.1109/ICME55011.2023.00404.

[12] F. Sun, P. Ren, B. Yin, F. Wang, H. Li, CATNet: A cascaded and aggregated transformer network for RGB-D salient object detection, IEEE Trans. Multimed. 26 (2024) 2249–2262, http://dx.doi.org/10.1109/TMM.2023.3294003.

[13] G. Liao, W. Gao, G. Li, J. Wang, S. Kwong, Cross-collaborative fusion-encoder network for robust RGB-thermal salient object detection, IEEE Trans. Circuits Syst. Video Technol. 32 (2022) 7646–7661, http://dx.doi.org/10.1109/TCSVT.2022.3184840.

[14] D. Wang, J. Liu, R. Liu, X. Fan, An interactively reinforced paradigm for joint infrared-visible image fusion and saliency object detection, Inf. Fusion 98 (2023) 101828, http://dx.doi.org/10.1016/j.inffus.2023.101828.

[15] Y. Pang, X. Zhao, L. Zhang, H. Lu, CAVER: Cross-modal view-mixed transformer for bi-modal salient object detection, IEEE Trans. Image Process. 32 (2023) 892–904, http://dx.doi.org/10.1109/TIP.2023.3234702.

[16] J. Zhang, X. Yu, A. Li, P. Song, B. Liu, Y. Dai, Weakly-supervised salient object detection via scribble annotations, in: IEEE Conf. Comput. Vis. Pattern Recognit., 2020, pp. 12543–12552, http://dx.doi.org/10.1109/CVPR42600.2020.01256.

[17] Y. Xu, X. Yu, J. Zhang, L. Zhu, D. Wang, Weakly supervised RGB-D salient object detection with prediction consistency training and active scribble boosting, IEEE Trans. Image Process. 31 (2022) 2148–2161, http://dx.doi.org/10.1109/TIP.2022.3151999.

[18] Y. Wei, S. Ji, Scribble-based weakly supervised deep learning for road surface extraction from remote sensing images, IEEE Trans. Geosci. Remote Sens. 60 (2021) 1–12, http://dx.doi.org/10.1109/TGRS.2021.3061213.

[19] Q. Chen, Y. Hong, Scribble2d5: Weakly-supervised volumetric image segmentation via scribble annotations, in: Int. Conf. Med. Image Comput. Comput. Assisted Intervention, 2022, pp. 234–243, http://dx.doi.org/10.1007/978-3-031-16452-1_23.

[20] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A.C. Berg, W.-Y. Lo, P. Dollár, R. Girshick, Segment anything, IEEE Int. Conf. Comput. Vis. (2023) 3992–4003, http://dx.doi.org/10.1109/ICCV51070.2023.00371.

[21] L. Wang, M. Zhang, W. Shi, CS-WSCDNet: Class activation mapping and segment anything model-based framework for weakly supervised change detection, IEEE Trans. Geosci. Remote Sens. 61 (2023) 1–12, http://dx.doi.org/10.1109/TGRS.2023.3330479.

[22] B. Zhou, A. Khosla, A. Lapedriza, A. Oliva, A. Torralba, Learning deep features for discriminative localization, in: IEEE Conf. Comput. Vis. Pattern Recognit., 2016, pp. 2921–2929, http://dx.doi.org/10.1109/CVPR.2016.319.

[23] Y. Wang, R. Wang, X. He, C. Lin, T. Wang, Q. Jia, X. Fan, WBNet: Weakly-supervised salient object detection via scribble and pseudo-background priors, Pattern Recognit. 154 (2024) 110579, http://dx.doi.org/10.1016/j.patcog.2024.110579.

[24] G. Shin, S. Albanie, W. Xie, Unsupervised salient object detection with spectral cluster voting, in: IEEE Conf. Comput. Vis. Pattern Recognit., 2022, pp. 3971–3980, http://dx.doi.org/10.1109/CVPRW56347.2022.00442.

[25] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, Ł. Kaiser, I. Polosukhin, Attention is all you need, in: Adv. Neural Inf. Process. Syst., 30, 2017, pp. 6000–6010.

[26] J. Hu, L. Shen, G. Sun, Squeeze-and-excitation networks, in: IEEE Conf. Comput. Vis. Pattern Recognit., 2018, pp. 7132–7141, http://dx.doi.org/10.1109/CVPR.2018.00745.

[27] C. He, K. Li, Y. Zhang, G. Xu, L. Tang, Y. Zhang, Z. Guo, X. Li, Weakly-supervised concealed object segmentation with sam-based pseudo labeling and multi-scale feature grouping, in: Adv. Neural Inf. Process. Syst., 2024.

[28] H. Kweon, K.-J. Yoon, From SAM to CAMs: Exploring segment anything model for weakly supervised semantic segmentation, in: IEEE Conf. Comput. Vis. Pattern Recognit., 2024, pp. 19499–19509, http://dx.doi.org/10.1109/CVPR52733.2024.01844.

[29] S. Liu, S. Zhi, E. Johns, A.J. Davison, Bootstrapping semantic segmentation with regional contrast, in: Int. Conf. Learn. Represent., 2022.

[30] H. Zhang, Y. Su, X. Xu, K. Jia, Improving the generalization of segmentation foundation model under distribution shift via weakly supervised adaptation, in: IEEE Conf. Comput. Vis. Pattern Recognit., 2024, pp. 23385–23395, http://dx.doi.org/10.1109/CVPR52733.2024.02207.

[31] Y. Liu, P. Wang, Y. Cao, Z. Liang, R.W. Lau, Weakly-supervised salient object detection with saliency bounding boxes, IEEE Trans. Image Process. 30 (2021) 4423–4435, http://dx.doi.org/10.1109/TIP.2021.3071691.

[32] H. Zhang, Y. Zeng, H. Lu, L. Zhang, J. Li, J. Qi, Learning to detect salient object with multi-source weak supervision, IEEE Trans. Pattern Anal. Mach. Intell. 44 (2021) 3577–3589, http://dx.doi.org/10.1109/TPAMI.2021.3059783.

[33] S. Gao, W. Zhang, Y. Wang, Q. Guo, C. Zhang, Y. He, W. Zhang, Weakly-supervised salient object detection using point supervision, in: AAAI Conf. Artif. Intell., 36, 2022, pp. 670–678, http://dx.doi.org/10.1609/aaai.v36i1.19947.

[34] S. Yu, B. Zhang, J. Xiao, E.G. Lim, Structure-consistent weakly supervised salient object detection with local saliency coherence, in: AAAI Conf. Artif. Intell., 35, 2021, pp. 3234–3242, http://dx.doi.org/10.1609/aaai.v35i4.16434.

[35] A. Li, Y. Mao, J. Zhang, Y. Dai, Mutual information regularization for weakly-supervised RGB-D salient object detection, IEEE Trans. Circuits Syst. Video Technol. 34 (2024) 397–410, http://dx.doi.org/10.1109/TCSVT.2023.3285249.

[36] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, A.L. Yuille, Deeplab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected crfs, IEEE Trans. Pattern Anal. Mach. Intell. 40 (2017) 834–848, http://dx.doi.org/10.1109/TPAMI.2017.2699184.

[37] K. Desingh, K.M. Krishna, D. Rajan, C. Jawahar, Depth really matters: Improving Visual Salient Region detection with depth., in: Brit. Mach. Vis. Conf., 2013, pp. 1–11.

[38] Y. Piao, Z. Rong, M. Zhang, W. Ren, H. Lu, A2dele: Adaptive and attentive depth distiller for efficient RGB-D salient object detection, in: IEEE Conf. Comput. Vis. Pattern Recognit., 2020, pp. 9060–9069, http://dx.doi.org/10.1109/CVPR42600.2020.00908.

[39] G. Chen, F. Shao, X. Chai, H. Chen, Q. Jiang, X. Meng, Y.-S. Ho, CGMDRNet: Cross-guided modality difference reduction network for RGB-T salient object detection, IEEE Trans. Circuits Syst. Video Technol. 32 (2022) 6308–6323, http://dx.doi.org/10.1109/TCSVT.2022.3166914.

[40] Z. Zhang, J. Wang, Y. Han, Saliency prototype for RGB-D and RGB-T salient object detection, in: ACM Int. Conf. Multimedia, 2023, pp. 3696–3705.

[41] X. Jiang, Y. Hou, H. Tian, L. Zhu, Mirror complementary transformer network for RGB-thermal salient object detection, IET Comput. Vis. 18 (1) (2024) 15–32, http://dx.doi.org/10.1049/cvi2.12221.

[42] J. Wang, G. Li, J. Shi, J. Xi, Weighted guided optional fusion network for rgb-t salient object detection, in: ACM Trans. Multimedia Comput. Commun. Appl., 20, 2024, pp. 1–20.

[43] W. Zhou, F. Sun, Q. Jiang, R. Cong, J.-N. Hwang, WaveNet: Wavelet network with knowledge distillation for RGB-T salient object detection, IEEE Trans. Image Process. 32 (2023) 3027–3039, http://dx.doi.org/10.1109/TIP.2023.3275538.

[44] K. Wang, D. Lin, C. Li, Z. Tu, B. Luo, Alignment-free RGBT salient object detection: Semantics-guided asymmetric correlation network and a unified benchmark, IEEE Trans. Multimed. (2024) 10692–10707, http://dx.doi.org/10.1109/TMM.2024.3410542.

[45] S. Ren, D. Zhou, S. He, J. Feng, X. Wang, Shunted self-attention via multi-scale token aggregation, in: IEEE Conf. Comput. Vis. Pattern Recognit., 2022, pp. 10853–10862, http://dx.doi.org/10.1109/CVPR52688.2022.01058.

[46] R. Tarjan, Depth-first search and linear graph algorithms, SIAM J. Comput. 1 (1972) 146–160, http://dx.doi.org/10.1109/SWAT.1971.10.

[47] A. Ali, H. Touvron, M. Caron, P. Bojanowski, M. Douze, A. Joulin, I. Laptev, N. Neverova, G. Synnaeve, J. Verbeek, et al., Xcit: Cross-covariance image transformers, in: Adv. Neural Inf. Process. Syst., 34, 2021, pp. 20014–20027.

[48] Z. Shen, M. Zhang, H. Zhao, S. Yi, H. Li, Efficient attention: Attention with linear complexities, in: IEEE Winter Conf. Appl. Comput. Vis., 2021, pp. 3530–3538, http://dx.doi.org/10.1109/WACV48630.2021.00357.

[49] J. Wei, S. Wang, Q. Huang, F^3net: fusion, feedback and focus for salient object detection, in: AAAI Conf. Artif. Intell., 34, 2020, pp. 12321–12328, http://dx.doi.org/10.1609/aaai.v34i07.6916.

[50] M. Tang, A. Djelouah, F. Perazzi, Y. Boykov, C. Schroers, Normalized cut loss for weakly-supervised cnn segmentation, in: IEEE Conf. Comput. Vis. Pattern Recognit., 2018, pp. 1818–1827, http://dx.doi.org/10.1109/CVPR.2018.00195.

[51] Z. Tu, Y. Ma, Z. Li, C. Li, J. Xu, Y. Liu, RGBT salient object detection: A large-scale dataset and benchmark, IEEE Trans. Multimed. 25 (2023) 4163–4176, http://dx.doi.org/10.1109/TMM.2022.3171688.

[52] G. Wang, C. Li, Y. Ma, A. Zheng, J. Tang, B. Luo, RGB-T saliency detection benchmark: Dataset, baselines, analysis and a novel approach, in: Image Graph. Technol. Appl., 2018, pp. 359–369, http://dx.doi.org/10.1007/978-981-13-1702-6_36.

[53] Z. Tu, T. Xia, C. Li, X. Wang, Y. Ma, J. Tang, RGB-T image saliency detection via collaborative graph learning, IEEE Trans. Multimed. 22 (2019) 160–173, http://dx.doi.org/10.1109/TMM.2019.2924578.

[54] Y. Cheng, H. Fu, X. Wei, J. Xiao, X. Cao, Depth enhanced saliency detection method, in: Int. Conf. Internet Multimedia Comput. Service, 2014, pp. 23–27.

[55] H. Peng, B. Li, W. Xiong, W. Hu, R. Ji, Rgbd salient object detection: a benchmark and algorithms, in: Eur. Conf. Comput. Vis., Springer, 2014, pp. 92–109, http://dx.doi.org/10.1007/978-3-319-10578-9_7.

[56] D.-P. Fan, Z. Lin, Z. Zhang, M. Zhu, M.-M. Cheng, Rethinking RGB-D salient object detection: Models, data sets, and large-scale benchmarks, IEEE Trans. Neural Netw. Learn. Syst. 32 (5) (2020) 2075–2089, http://dx.doi.org/10.1109/TNNLS.2020.2996406.

[57] Y. Niu, Y. Geng, X. Li, F. Liu, Leveraging stereopsis for saliency analysis, in: IEEE Conf. Comput. Vis. Pattern Recognit., 2012, pp. 454–461, http://dx.doi.org/10.1109/CVPR.2012.6247708.

[58] C. Zhu, G. Li, A three-pathway psychobiological framework of salient object detection using stereoscopic technology, in: IEEE Int. Conf. Comput. Vis., 2017, pp. 3008–3014, http://dx.doi.org/10.1109/ICCVW.2017.355.

[59] N. Li, J. Ye, Y. Ji, H. Ling, J. Yu, Saliency detection on light field, IEEE Trans. Pattern Anal. Mach. Intell. 39 (2017) 1605–1616, http://dx.doi.org/10.1109/TPAMI.2016.2610425.

[60] D.-P. Fan, M.-M. Cheng, Y. Liu, T. Li, A. Borji, Structure-measure: A new way to evaluate foreground maps, in: IEEE Int. Conf. Comput. Vis., 2017, pp. 4548–4557, http://dx.doi.org/10.1109/ICCV.2017.487.

[61] R. Achanta, S. Hemami, F. Estrada, S. Susstrunk, Frequency-tuned salient region detection, in: IEEE Conf. Comput. Vis. Pattern Recognit., 2009, pp. 1597–1604, http://dx.doi.org/10.1109/CVPR.2009.5206596.

[62] D.-P. Fan, C. Gong, Y. Cao, B. Ren, M.-M. Cheng, A. Borji, Enhanced-alignment measure for binary foreground map evaluation, in: Int. Joint Conf. Artif. Intell., 2018, pp. 698–704.

[63] R. Margolin, L. Zelnik-Manor, A. Tal, How to evaluate foreground maps? in: IEEE Conf. Comput. Vis. Pattern Recognit., 2014, pp. 248–255, http://dx.doi.org/10.1109/CVPR.2014.39.

[64] F. Perazzi, P. Krähenbühl, Y. Pritch, A. Hornung, Saliency filters: Contrast based filtering for salient region detection, in: IEEE Conf. Comput. Vis. Pattern Recognit., IEEE, 2012, pp. 733–740, http://dx.doi.org/10.1109/CVPR.2012.6247743.

[65] A. Borji, M.-M. Cheng, H. Jiang, J. Li, Salient object detection: A benchmark, IEEE Trans. Image Process. 24 (2015) 5706–5722, http://dx.doi.org/10.1109/TIP.2015.2487833.

[66] O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein, ImageNet large scale visual recognition challenge, Int. J. Comput. Vis. 115 (2015) 211–252, http://dx.doi.org/10.1007/s11263-015-0816-y.

[67] S. Bock, M. Weiß, A proof of local convergence for the adam optimizer, in: Int. Joint Conf. Neural Networks, IEEE, 2019, pp. 1–8, http://dx.doi.org/10.1109/IJCNN.2019.8852239.

[68] Z. Liu, M. Hayat, H. Yang, D. Peng, Y. Lei, Deep hypersphere feature regularization for weakly supervised RGB-D salient object detection, IEEE Trans. Image Process. 32 (2023) 5423–5437, http://dx.doi.org/10.1109/TIP.2023.3318953.

[69] Y. Wang, L. Zhang, P. Zhang, Y. Zhuge, J. Wu, H. Yu, H. Lu, Learning local-global representation for scribble-based RGB-D salient object detection via transformer, IEEE Trans. Circuits Syst. Video Technol. (2024) 11592–11604, http://dx.doi.org/10.1109/TCSVT.2024.3424651.

[70] L. Li, J. Han, N. Liu, S. Khan, H. Cholakkal, R.M. Anwer, F.S. Khan, Robust perception and precise segmentation for scribble-supervised RGB-D saliency detection, IEEE Trans. Pattern Anal. Mach. Intell. 46 (2024) 479–496, http://dx.doi.org/10.1109/TPAMI.2023.3324807.

[71] Z. Tu, Z. Li, C. Li, J. Tang, Weakly alignment-free RGBT salient object detection with deep correlation network, IEEE Trans. Image Process. 31 (2022) 3752–3764, http://dx.doi.org/10.1109/TIP.2022.3176540.

[72] N. Nayal, M. Yavuz, J.F. Henriques, F. Güney, Rba: Segmenting unknown regions rejected by all, in: IEEE Int. Conf. Comput. Vis., 2023, pp. 711–722, http://dx.doi.org/10.1109/ICCV51070.2023.00072.